

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	SWTID-2026-6491
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

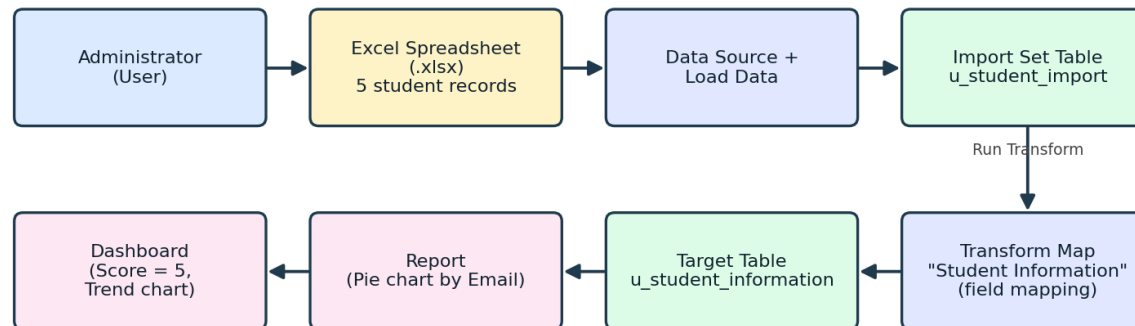
Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Architecture: Import Data using Transform Maps (Spreadsheet) on ServiceNow

<https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Architecture - ServiceNow Spreadsheet Import with Transform Map



Transform history: Total 5 | Inserts 5 | Updates 0 | Ignored 0 | Skipped 0 | Errors 0

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Administrator uses the ServiceNow web UI to create the table, upload the spreadsheet and view reports/dashboards	ServiceNow Web UI, Microsoft Excel (.xlsx)
2.	Application Logic-1	Data Source and Load Data: spreadsheet is loaded into the import set table u_student_import	ServiceNow Import Sets, Data Source
3.	Application Logic-2	Transform Map "Student Information" maps source fields to target fields and inserts the records	ServiceNow Transform Map, Field Maps
4.	Application Logic-3	Report and Dashboard: pie chart report by Email, single-score and line-chart widgets	ServiceNow Reports and Dashboards
5.	Database	Tables u_student_import and u_student_information; fields Student ID (Integer), Student Name, Department, Email (String), Year (Integer)	ServiceNow relational database
6.	Cloud Database	Database hosted with the cloud-based ServiceNow instance	ServiceNow Cloud Instance

7.	File Storage	Spreadsheet stored as an attachment on the Data Source record	ServiceNow Attachments (sys_attachment)
8.	External API-1	Not used in this project	N/A
9.	External API-2	Not used in this project	N/A
10.	Machine Learning Model	Not applicable to this project	N/A
11.	Infrastructure (Server / Cloud)	Application deployment on cloud. Local system: web browser only. Cloud server configuration: ServiceNow-hosted instance	ServiceNow Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Built on the ServiceNow low-code platform; no open-source framework needed	ServiceNow platform
2.	Security Implementations	Role-based access control and Application Access settings on the custom table	ServiceNow ACLs, Roles
3.	Scalable Architecture	Multi-tier cloud platform; the same Transform Map can be reused for larger files	ServiceNow multi-tier architecture
4.	Availability	Managed cloud hosting provided by ServiceNow	ServiceNow Cloud
5.	Performance	Import Set processing inserts 5 records in about 1 second	Import Sets, Transform Map

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>